

Ethanol Blending in Gasoline - Bolivia

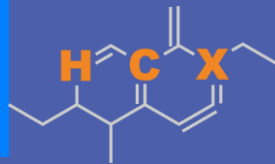
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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



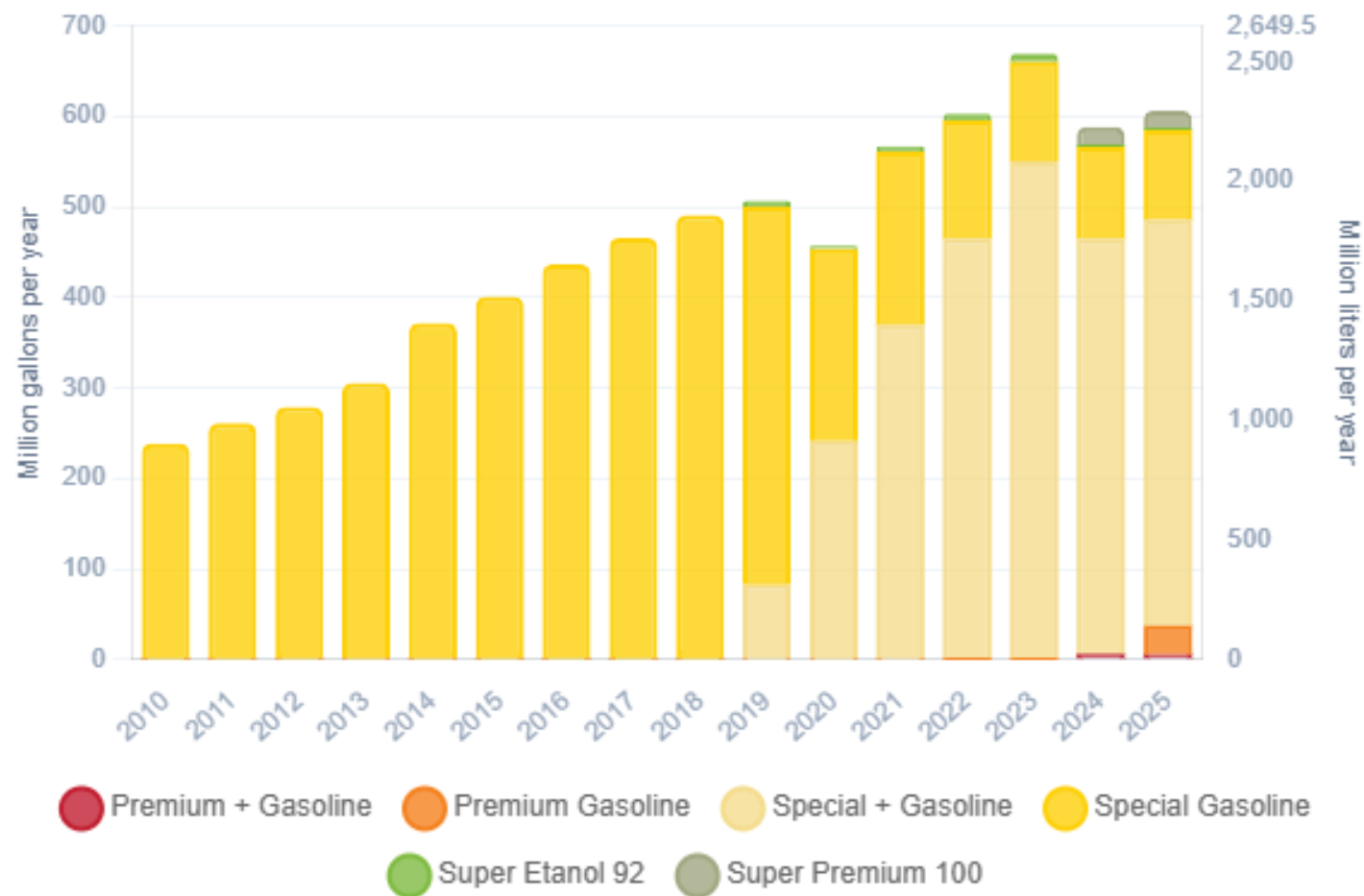
Ethanol Blending in Gasoline - Bolivia



In 2025, gasoline demand was 606 million gallons (2,295 million liters). Gasoline Imports represented 52% of demand, and are supplied from Brazil, Chile, Argentina, and the United States. Regular Gasoline (Especial) with 85 RON Octane represented 90% of demand and the Gasoline with a 95 RON or more octane, the remaining 10%. Gasoline production has decreased in the last years.

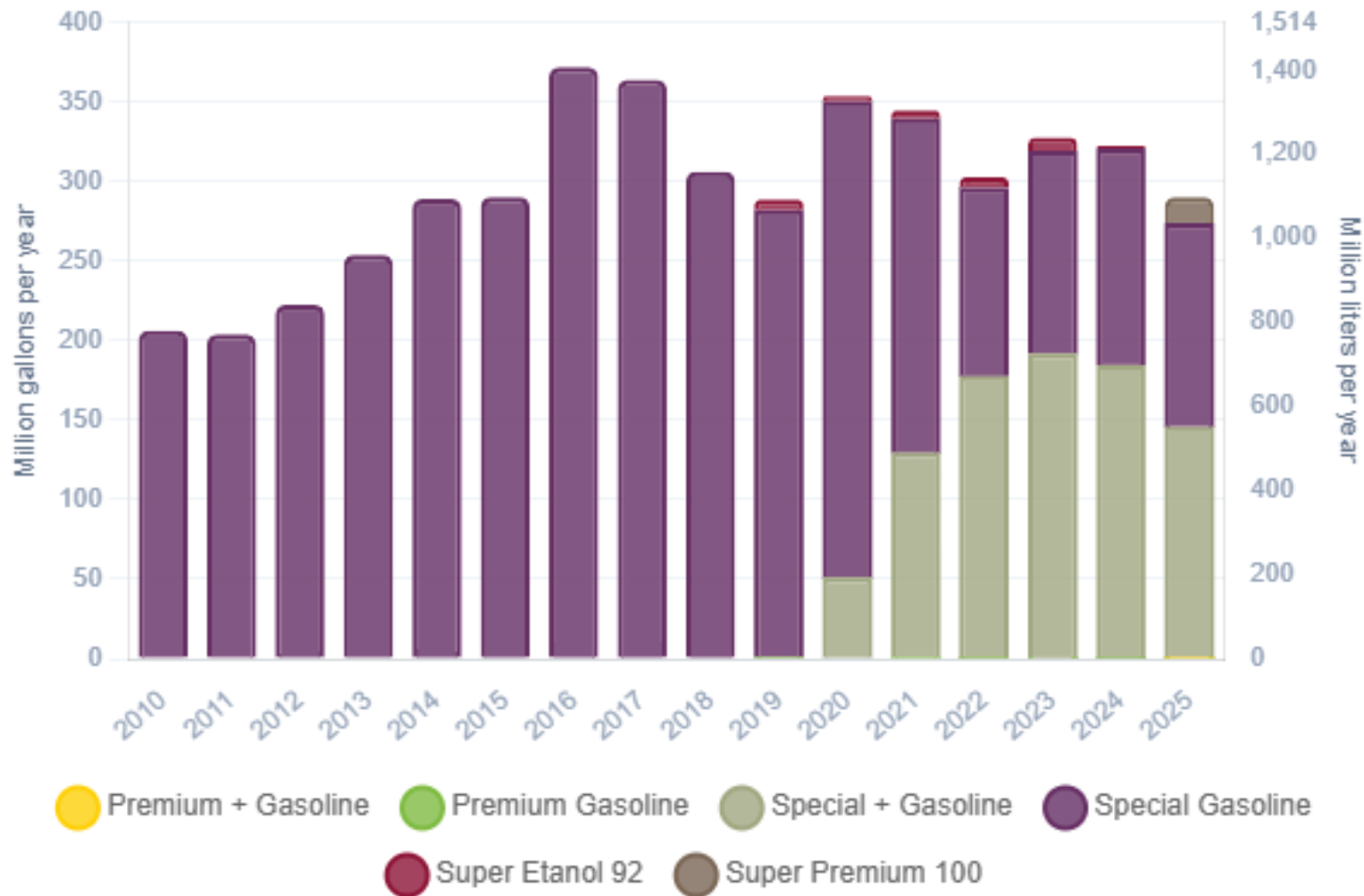
Gasoline E8 and E12 are currently marketed (Decreto Supremo No. 4718:ANH), but recently was approved blends of 25%. About 80% of gasoline demand contains ethanol. All ethanol consumed in Bolivia is produced by five local companies: Azúcar Aguaí, Unagro, Poplar Capital, Granosol, and Guabirá.

Gasoline Demand in Bolivia



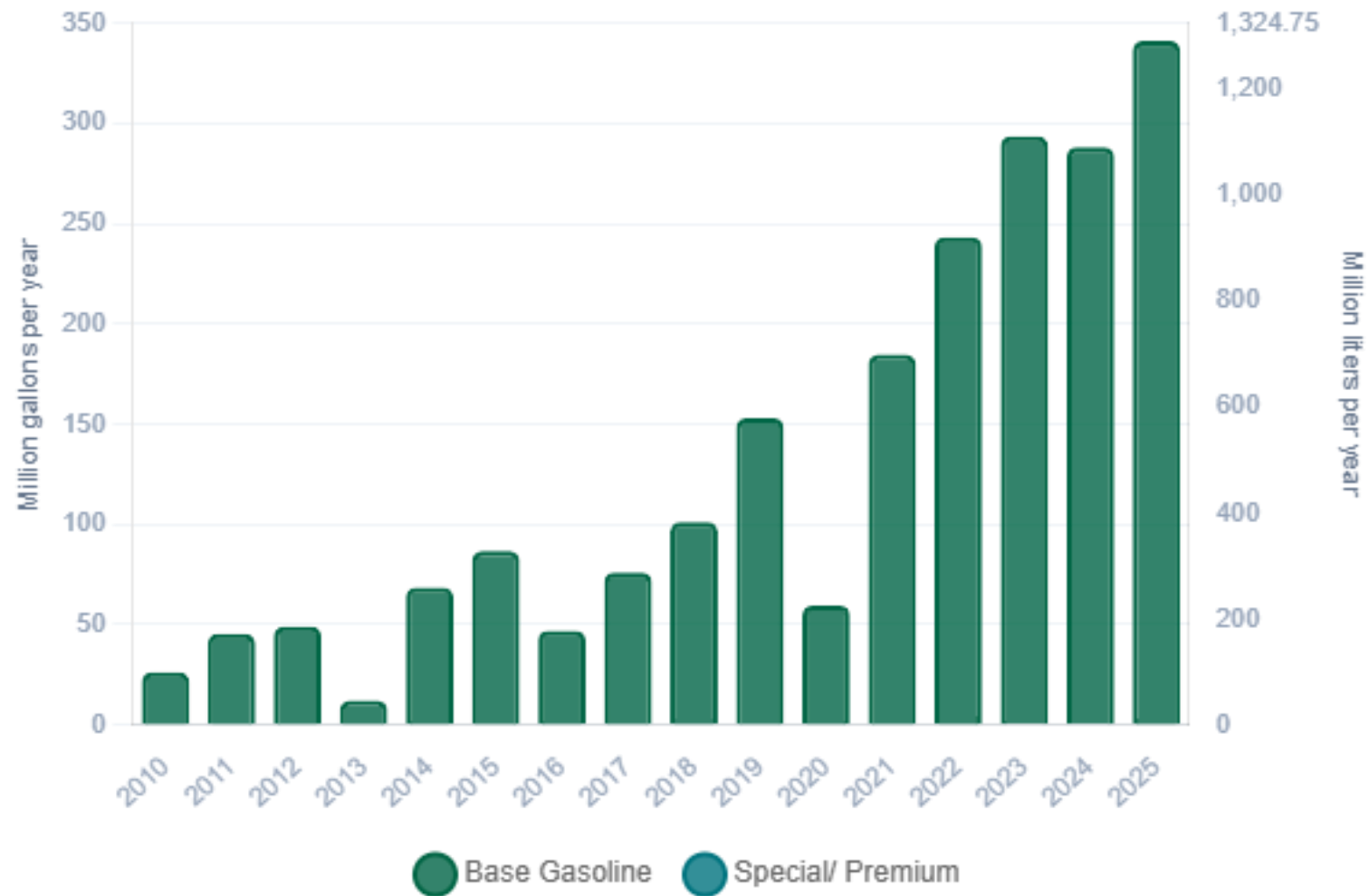
Source: ANH,INE - Bolivia

Gasoline Production in Bolivia



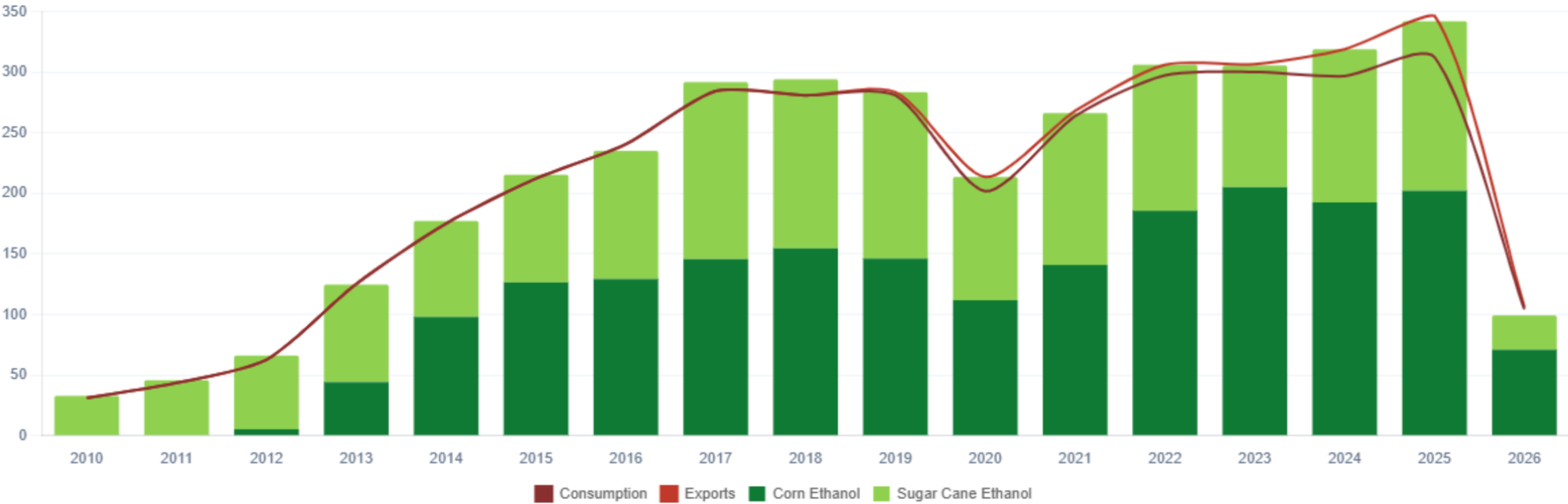
Source: ANH,INE - Bolivia

Gasoline Imports in Bolivia



Source: ANH,INE - Bolivia

Ethanol Balance in Bolivia



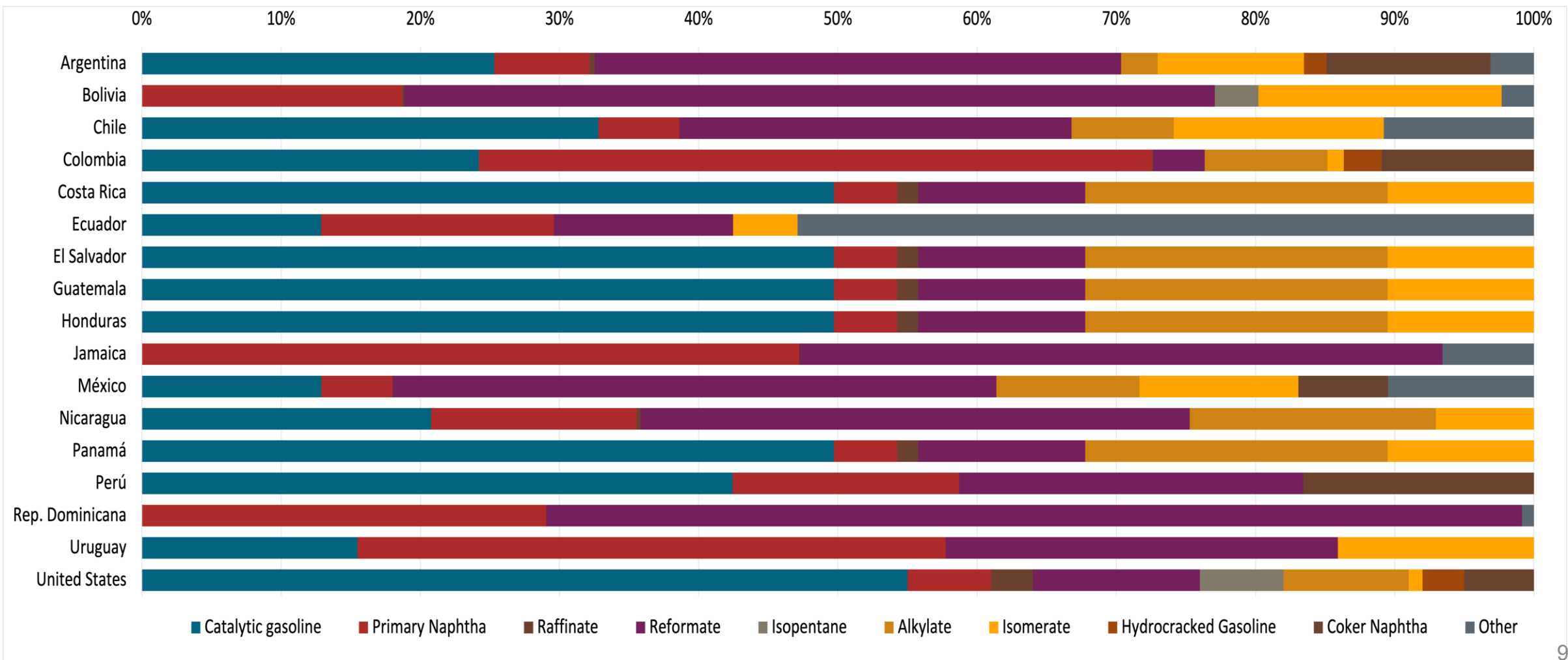
Source: ANH,INE - Bolivia

Gasoline Quality in Bolivia

Name	Supreme Decree 4718		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2022		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Regular	Gasoline Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 3%v/v (2,7%v/v for E12 in summer)	< 3%v/v (2,7%v/v for E12 in summer)	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 42 % v/v	< 48 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 18 mg/l	< 18 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 85	> 95	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2,7 %m/m	< 2,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 12 %v/v	< 12 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	⇔ 7-9 psi	⇔ 7-9 psi	⇔ 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	⇔ 7-9.5 psi	⇔ 7-9.5 psi				
RVP 37.8°C (Transition)	⇔ 7-9 psi	⇔ 7-9 psi				
MTBE			-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



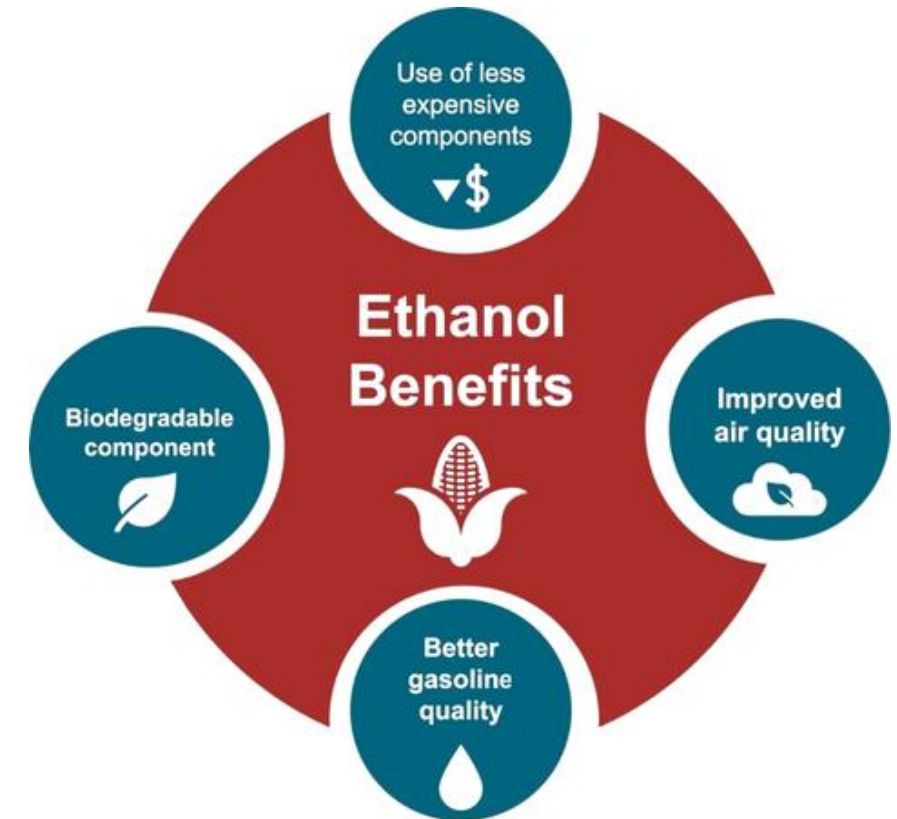
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

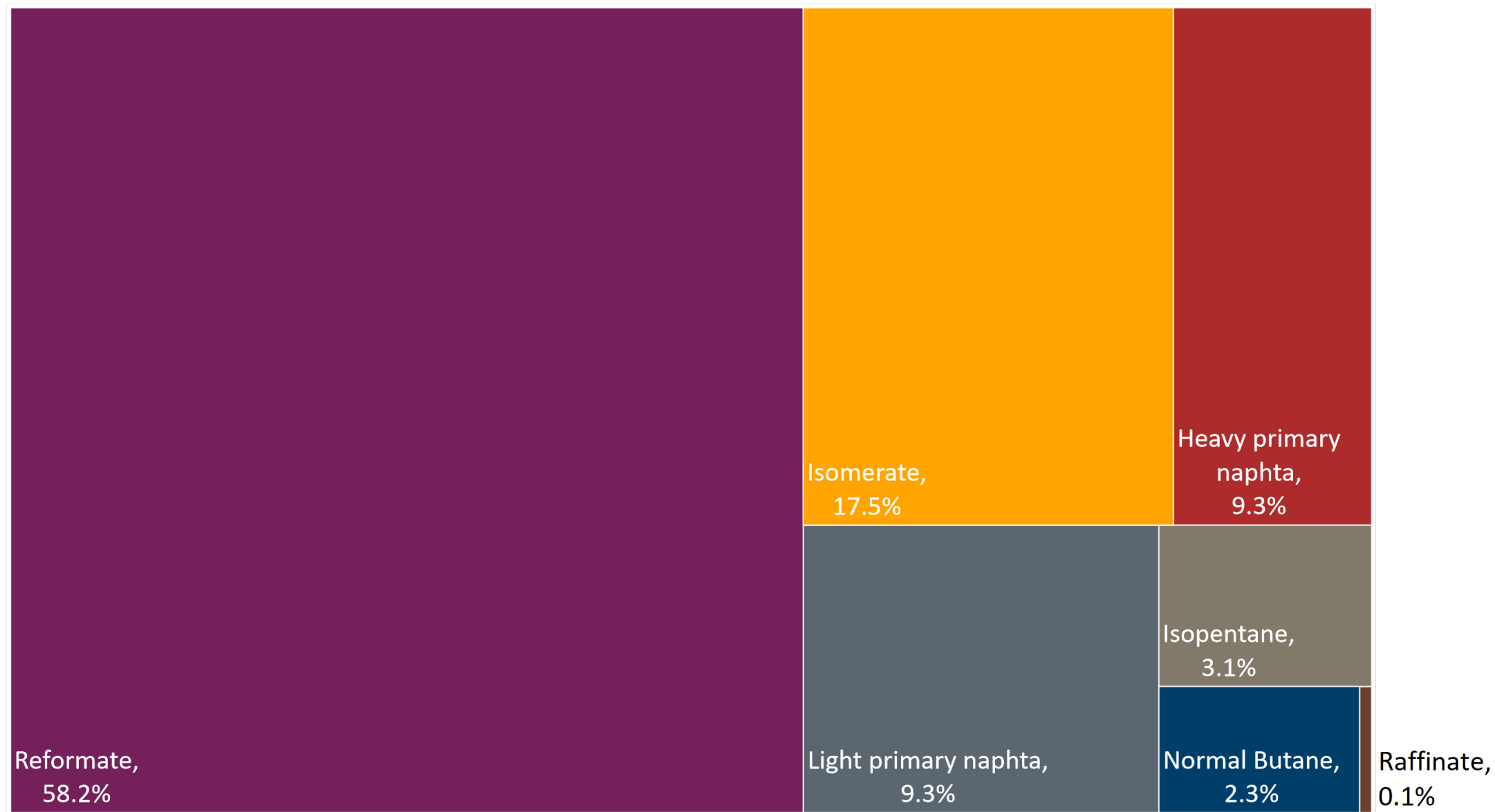
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

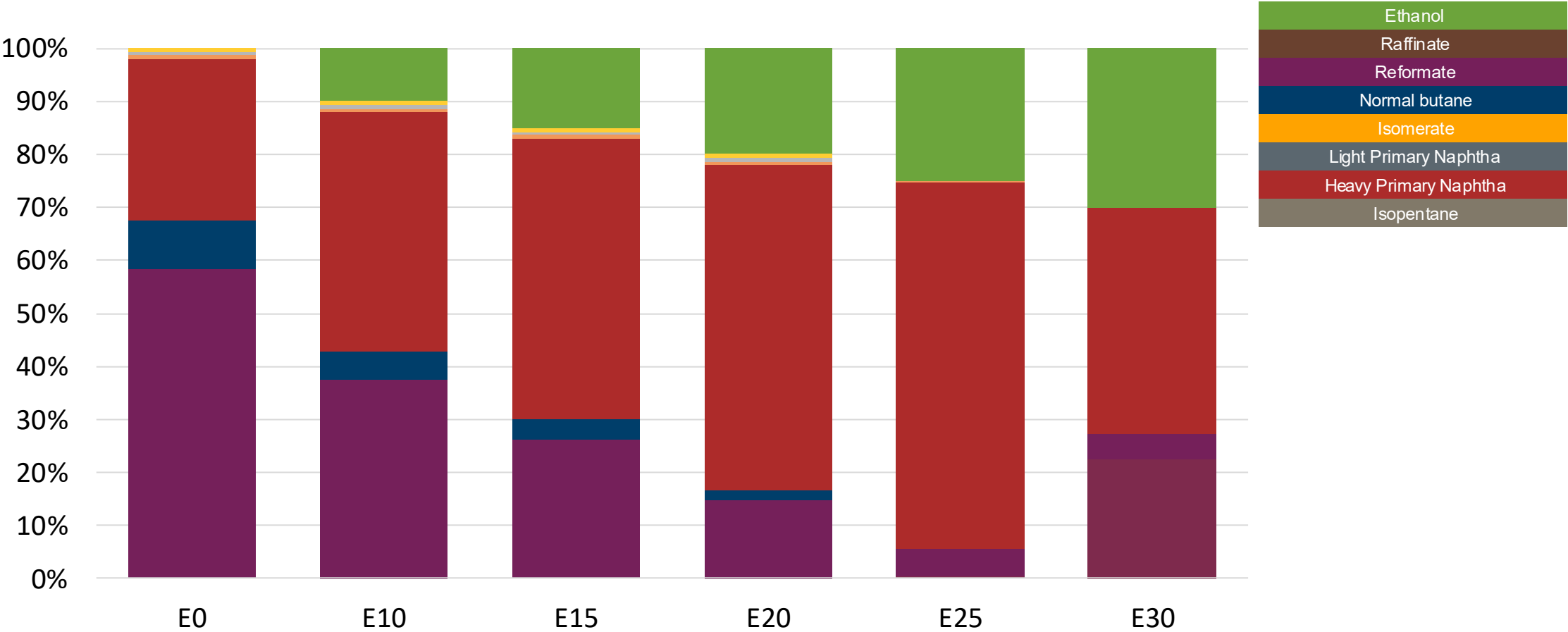
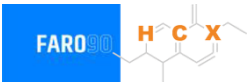
The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Available Blending Components

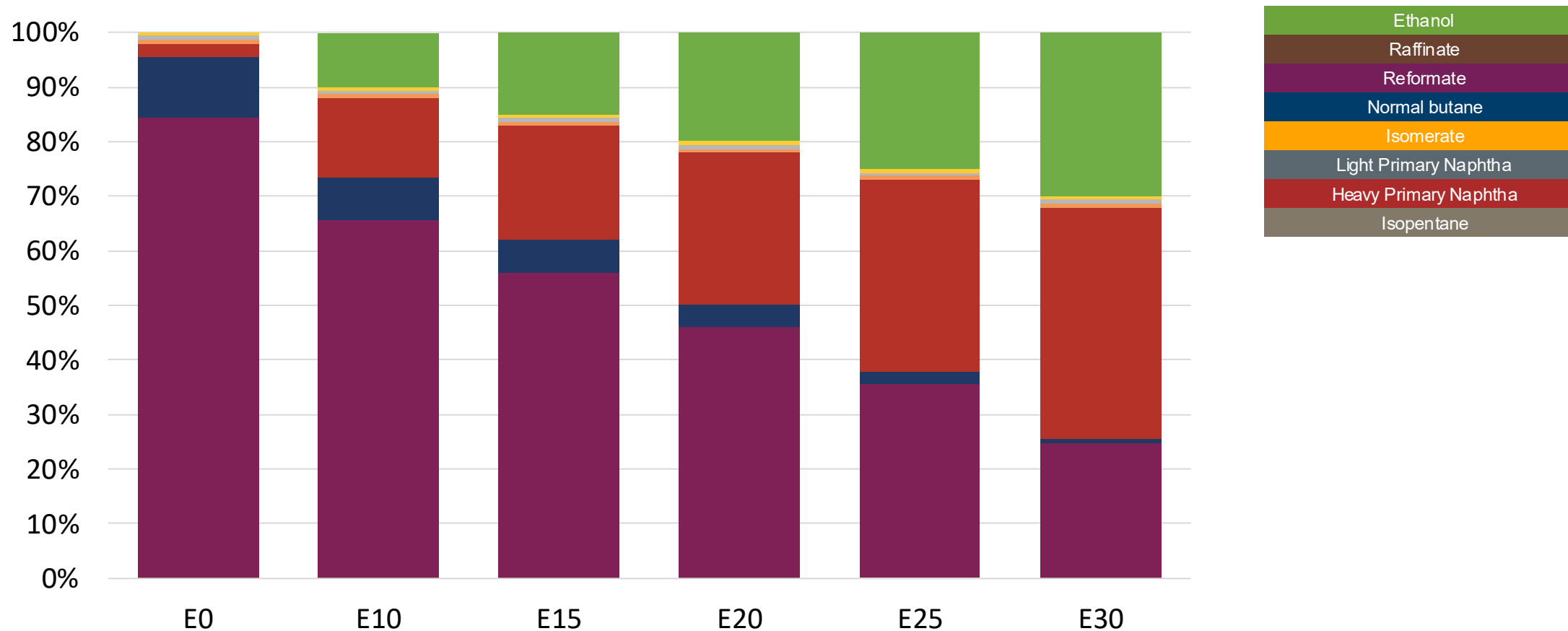


Ethanol Blending - Gasoline Especial – Constant Octane



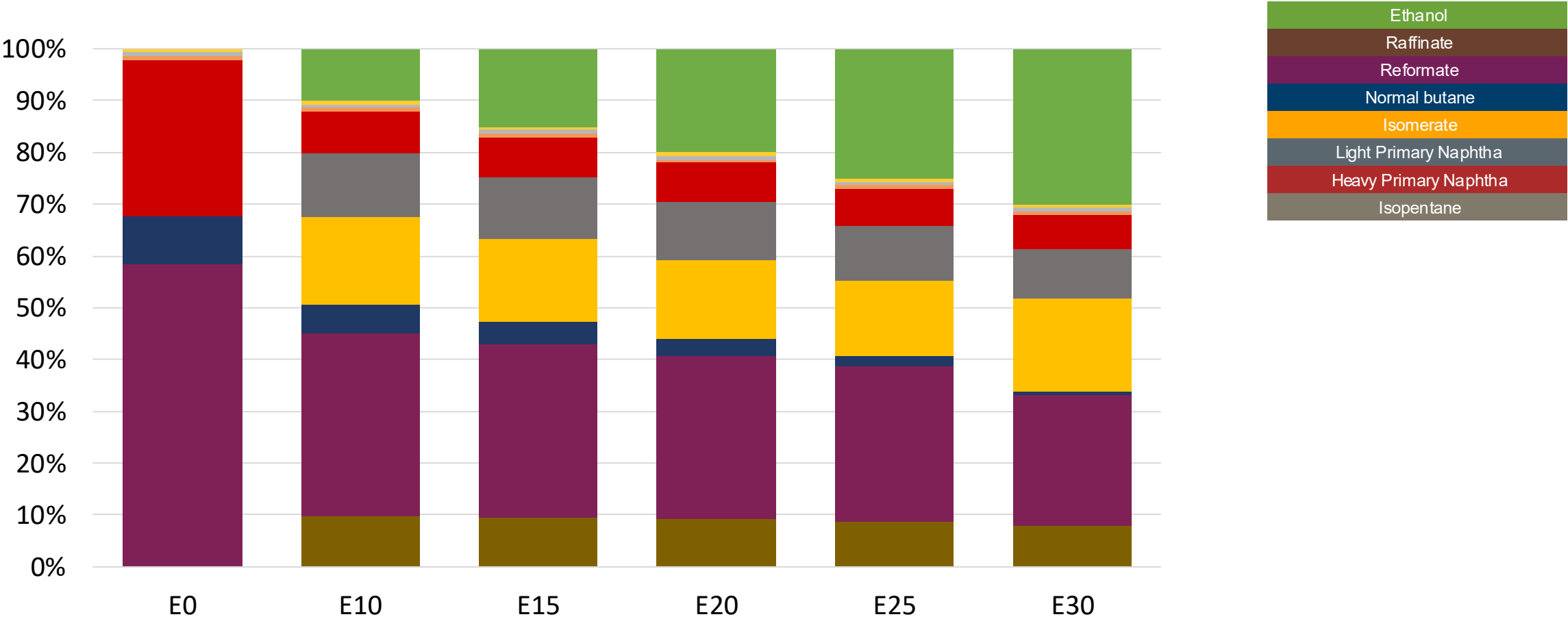
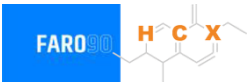
Octane (RON)	85.0	85.0	85.0	85.0	85.0	89.2
Price (US\$/gal)	\$2.05	\$1.75	\$1.58	\$1.41	\$1.23	\$1.33

Ethanol Blending – Gasoline Premium – Constant Octane



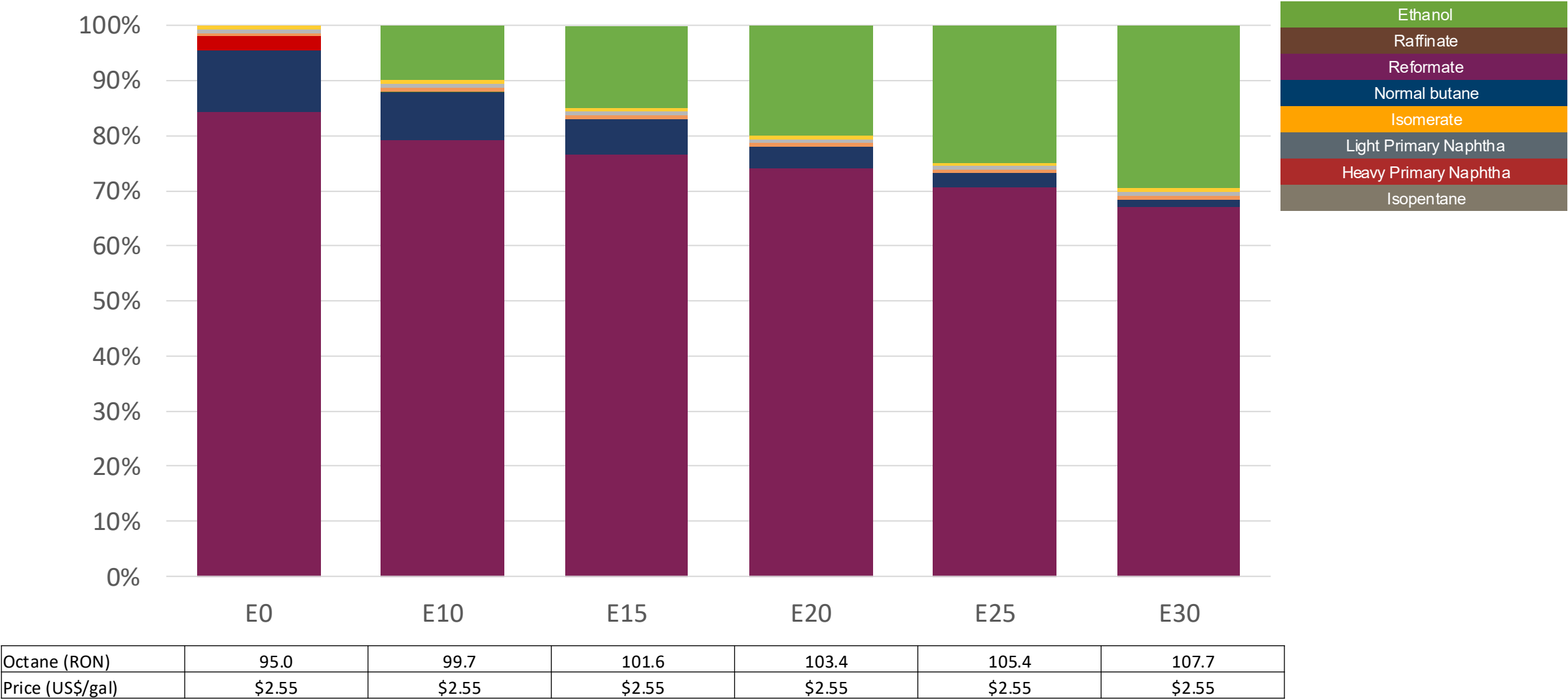
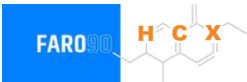
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US\$/gal)	\$2.55	\$2.29	\$2.16	\$2.01	\$1.86	\$1.71

Ethanol Blending - Gasoline Especial – Octane Increment



Octane (RON)	85.0	90.2	92.8	95.3	97.7	99.9
Price (US\$/gal)	\$2.05	\$2.05	\$2.05	\$2.05	\$2.05	\$2.05

Ethanol Blending - Gasoline Premium – Octane Increment



Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

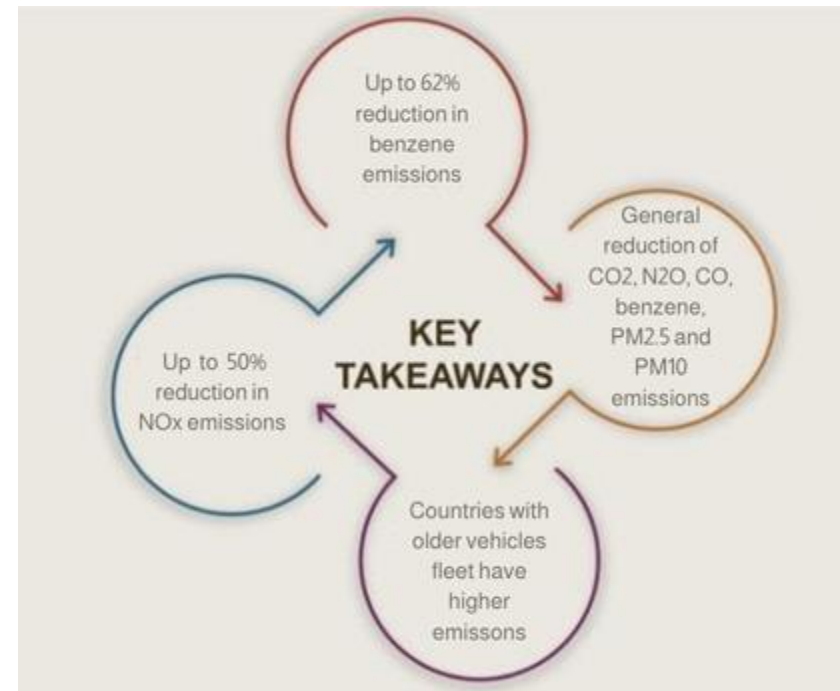
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet Bolivia

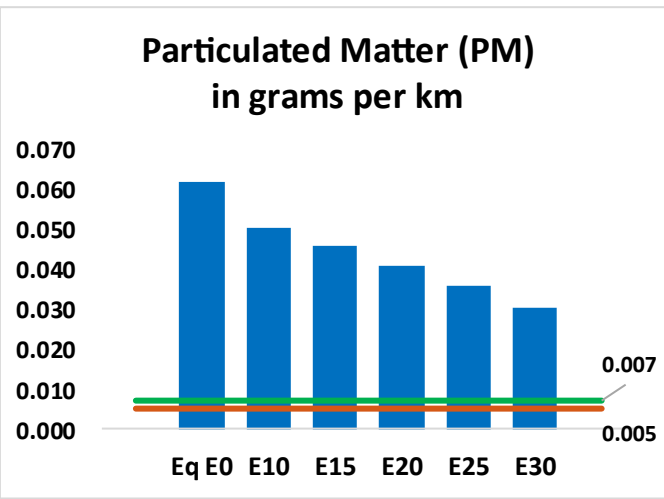
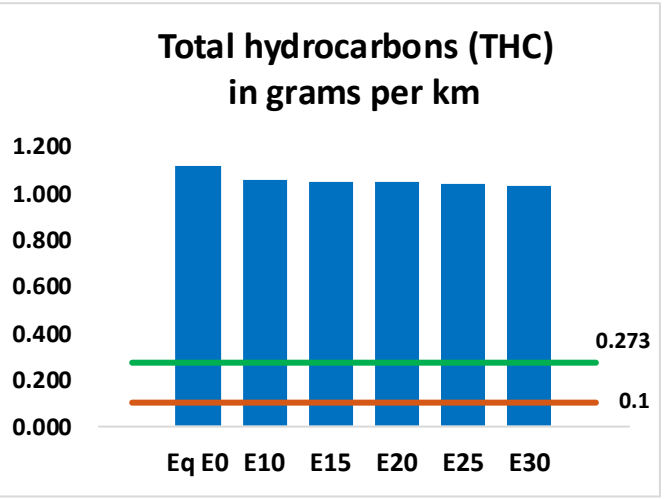
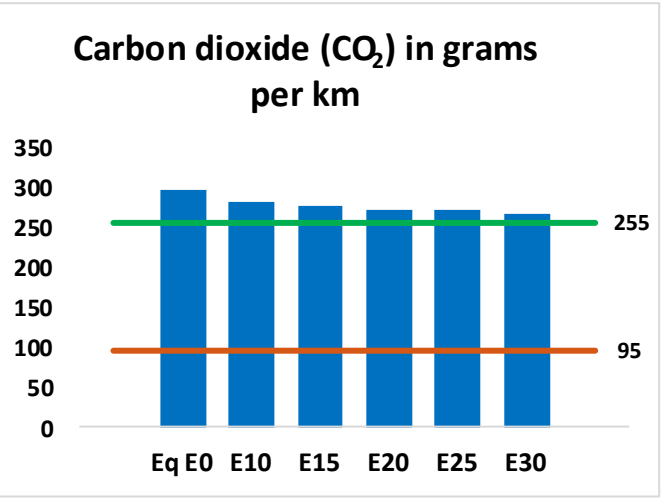
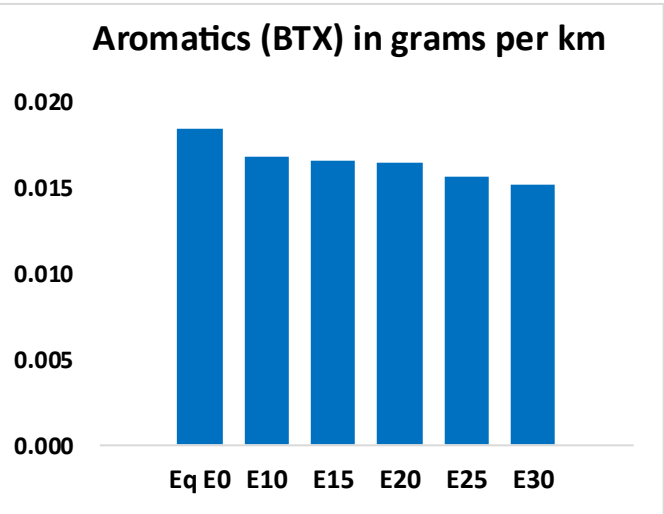
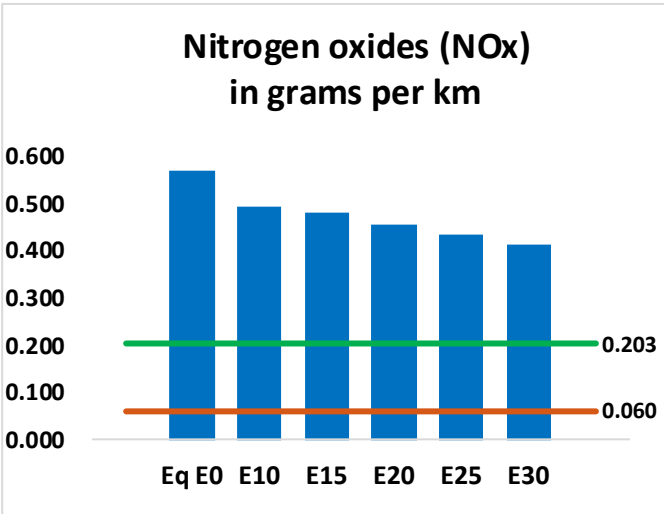
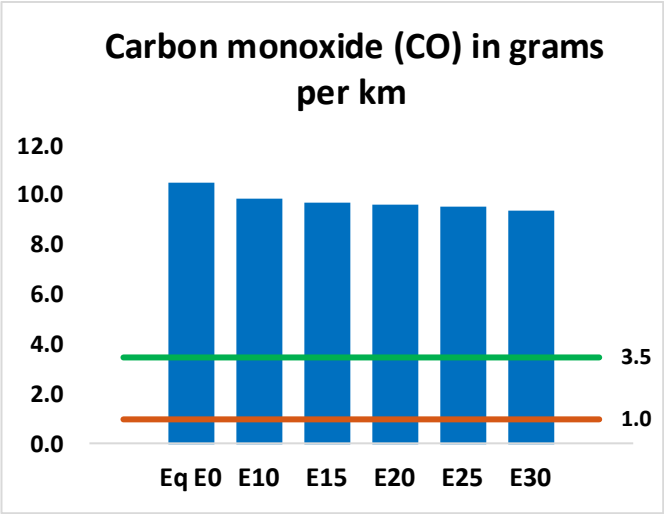
Vehicle Type	Engine	Exhaust	Recovery	Age	Number of vehicles			
Auto/Sml Truck	Multi FI-Pt	Euro V	PCV Tank	< 4 years, < 80 tkm	161,820			
				4-8 years, 80-160 tkm	228,679			
				> 8 years, > 160 tkm	177,284			
		Euro IV		> 8 years, > 160 tkm	177,284			
		Euro III		> 8 years, > 160 tkm	168,420			
		3-Ways		> 8 years, > 160 tkm	363,432			
Sml Engine	FI 4-cycle	Euro III	PCV	< 4years, < 80 tkm	294,897			
				4-8 years, 80-160 tkm	270,613			
		Euro II		4-8 years, 80-160 tkm	60,633			
				> 8 years, > 160 tkm	296,030			
Truck	FI	Euro V	PCV	< 4 years, < 80 tkm	33,765			
				4-8 years, 80-160 tkm	53,667			
				> 8 years, > 160 tkm	42,421			
		Euro IV		> 8 years, > 160 tkm	46,279			
		Euro III		> 8 years, > 160 tkm	26,996			
		3-Ways		> 8 years, > 160 tkm	269,955			

Vehicular Fleet: **2,672,176**
Gasoline Vehicle Fleet: 2,191,184
 Average Age: 12 years
 Motorcides: **34.5%**

Source: Instituto Nacional de Estadística – Bolivia (INE), analysis Faro 90

Bolivia – Vehicular Emissions

United States
Euro 6



Bolivia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	574,534	557,512	540,490	531,838	525,569	521,485	513,437	-6%	-9%
CO2	16,226,876	15,809,614	15,415,533	15,105,599	14,952,232	14,925,126	14,650,013	-5%	-8%
VOC	60,984	59,521	58,057	57,487	57,203	57,072	56,589	-5%	-6%
NOx	31,229	28,834	27,169	26,309	24,936	23,854	22,583	-13%	-20%
SOx	188	174	160	148	136	123	110	-15%	-28%
NH3	3,527	3,492	3,457	3,473	3,513	3,540	3,563	-2%	0%
N2O	835	830	827	831	848	858	867	-1%	2%
CH4	13,854	13,854	13,854	14,061	14,366	14,629	14,879	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	243	236	228	224	221	219	216	-6%	-9%
Acetaldehyde	662	760	1,115	1,698	2,313	2,643	3,123	68%	249%
Formaldehyde	2,700	2,625	3,060	3,593	3,764	4,164	4,534	13%	39%
Benzene	1,011	972	921	906	899	859	832	-9%	-11%
PM2.5	1,188	1,080	972	877	784	684	580	-18%	-34%
PM10	2,205	2,005	1,804	1,628	1,455	1,270	1,077	-18%	-34%

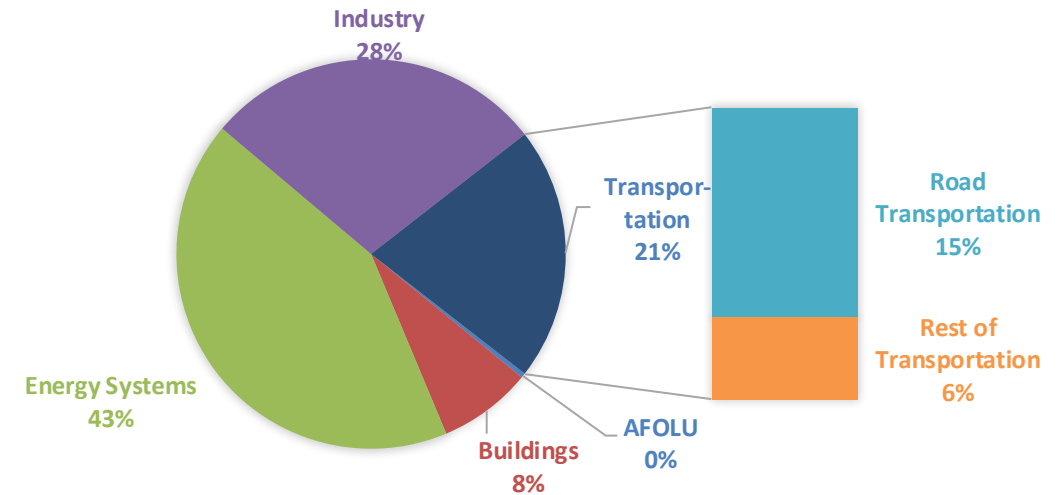
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

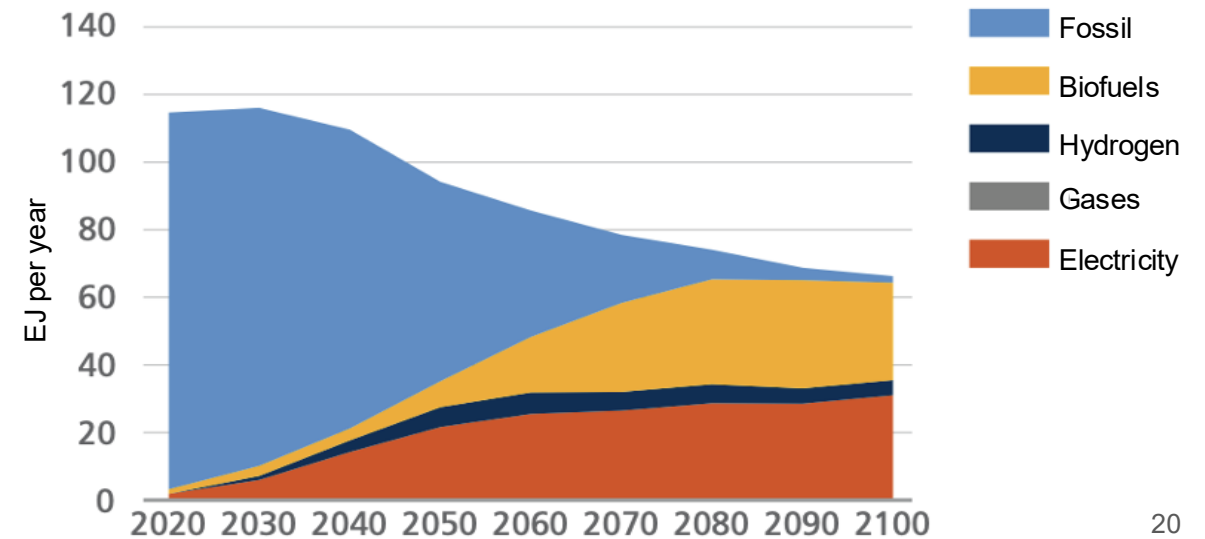
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

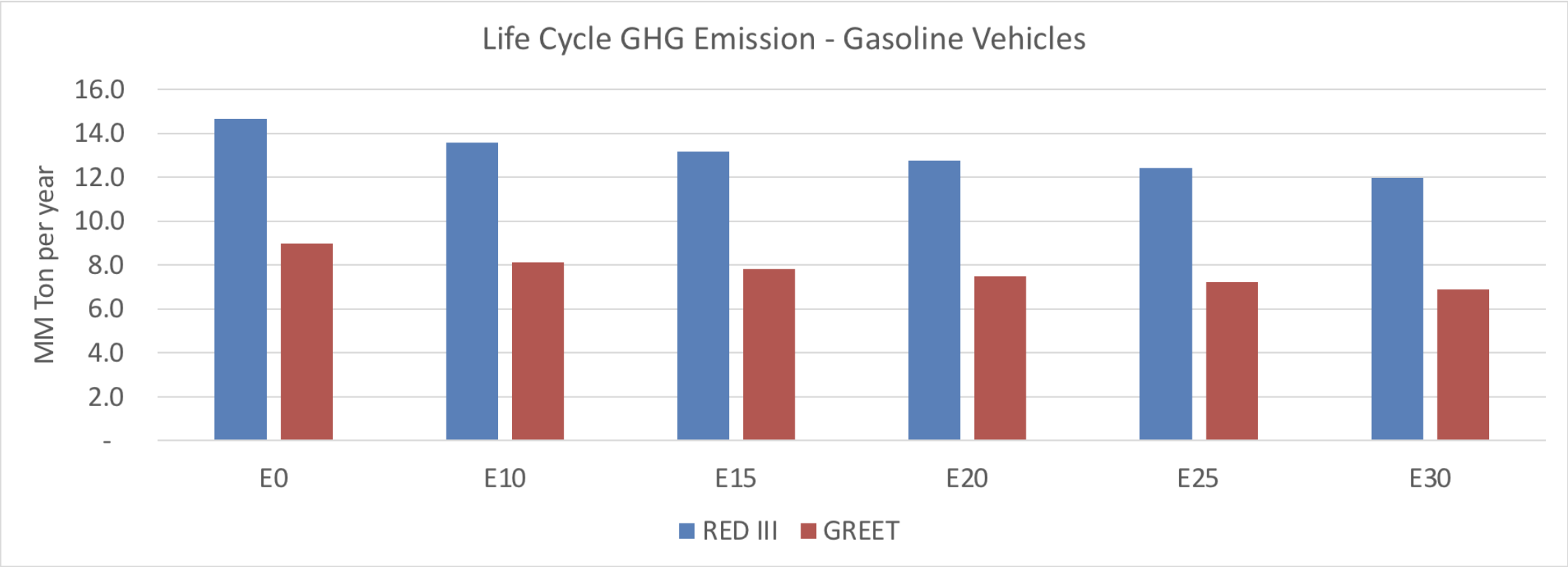
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Bolivia – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	63.8
2035 Target (MMT/year)	44.6
Delta	19.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.8%	13.2%	16.7%	19.5%	23.4%
RED II/III	0.0%	7.4%	10.2%	13%	15.3%	18.3%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	4.6%	6.2%	7.9%	9.2%	11.0%
RED II/III	0.0%	5.7%	7.8%	9.9%	11.7%	14.0%